Prova 2 - Transformadas de Laplace e Não homogêneas

① $y'' - 5y' = x - 2$, $y(0) = 0$, $y'(0) = 2$

① Achar y_c (complementar) a partir da eq. característica

$r^2 - 5r = 0$

$r^2 - 5r = 0$ $\quad \Delta = b^2 - 4ac$

$\Delta = (-5)^2 - 4(1)(0) = 25$

$r_1 = \frac{5 + \sqrt{25}}{2} = \frac{5+5}{2} = 5$

$r_2 = \frac{5-5}{2} = \frac{0}{2} = 0$

$y_c = c_1 e^{r_1 x} + c_2 e^{r_2 x} = c_1 e^{5x} + c_2 e^{0x} = c_1 e^{5x} + c_2$

② Encontrar y_p que satisfaça $y'' - 5y' = x - 2$

$y_p = Ax^2 + Bx \quad y_p' = 2Ax + B \quad y'' = 2A$

$y_p'' - 5y_p' = x - 2$

$2A - 5(2Ax + B) = x - 2$

$2A - 10Ax - 5B = x - 2$

$(-10A) \times (+2A - 5B) = 1x - 2$

$\begin{cases} -10A = 1 \rightarrow A = -\frac{1}{10} \\ 2A - 5B = -2 \end{cases}$

$2 \cdot \frac{-1}{10} - 5B = -2 \rightarrow -5B = -2 + \frac{1}{5} \rightarrow$

$-25B = -10 + 1 \rightarrow -25B = -9 \rightarrow B = \frac{9}{25}$

$\frac{-25B}{5}$

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$y_p = Ax^2 + Bx$, com $A = -\frac{1}{10}$ e $B = \frac{9}{25}$

$y_p = -\frac{1}{10}x^2 + \frac{9}{25}x$

③ Solução geral é $y = y_c + y_p = c_1 e^{5x} + c_2 - \frac{x^2}{10} + \frac{9x}{25}$

Resolvendo o P.V.I: $y(0) = 0$, $y'(0) = 2$

$y(0) = c_1 e^{5.0} + c_2 - \frac{0^2}{10} + \frac{9.0}{25}$

$e^{5.0} = 1$

$0 = c_1 + c_2$

$y' = f' + g' = 5c_1 e^{5x} - \frac{x}{5} + \frac{9}{25}$

$y'(0) = 5c_1 e^{5.0} - \frac{0}{5} + \frac{9}{25}$

$2 = 5c_1 + \frac{9}{25} \rightarrow 50 = 125c_1 + 9$

$\rightarrow 50 - 9 = 125c_1 \rightarrow \boxed{c_1 = \frac{41}{125}}$

$0 = c_1 + c_2 \rightarrow 0 = \frac{41}{125} + c_2 \rightarrow \boxed{c_2 = -\frac{41}{125}}$

$\boxed{y = \frac{41}{125}e^{5x} - \frac{41}{125} - \frac{x^2}{10} + \frac{9x}{25}}$

$(c_1 e^{5x})' = c_1 e^{5x} \cdot 5$

$(c_2)' = 0$

$\left(\frac{x^2}{10}\right)' = \frac{2x}{10} = \frac{x}{5}$

$\left(\frac{9x}{25}\right)' = \frac{9}{25}$

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